LECTURE 12: MARTINGALES - 2

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to `elre@princeton.edu`, thanks in advance!

E. Rebrova

Stopping times

Question: what if instead of running a stochastic process for as long as possible (to get to the limiting behavior), we would rather follow a random process for some time and stop it early, at some "good moment" to stop?

Definition 12.1. We say that a function $\tau : \Omega \to \{0, 1, 2, \ldots\} \cup \{+\infty\}$ is a **stopping time** if $\{\tau \leq n\} \in \mathcal{F}_n$ for all n.

Examples.

(1) $\tau = \inf\{n \geq 0 : X_n \in B\}$ (where B is a subset of the state space, e.g., $B \subseteq \mathcal{B}(\mathbb{R})$). Why?

$$\{\tau \leq n\} = \bigcup_{k \leq n} \{X_k \in B\} \in \sigma\left(\bigcup_{k \leq n} F_k\right) \subseteq \mathcal{F}_n.$$

(2) If σ and τ are stopping times, then (i) $\sigma \wedge \tau := \min\{\sigma, \tau\}$; (ii) $\sigma \vee \tau := \max\{\sigma, \tau\}$; and (iii) $\sigma + \tau$ are stopping times. Why? Exercise!

Non-examples.

(1) $\tau = \sup\{n \leq N : X_n \in B\}$.
(2) $\tau = \inf\{n \geq 0 : X_{n+1} - X_n < 0\}$ ("sell before the price drops").
(3) $\tau = \inf\{n \geq 0 : X_n \in B\} - c$ ("constant time before a stopping time").

Let $X_1, X_2, \ldots$ be i.i.d. random variables and N be a random time. When is $\mathbb{E}[\sum_{i=1}^N X_i] = \mathbb{E}[X_1] \cdot \mathbb{E}[N]$?

- It is not always true: for example, consider $X_1, X_2, \ldots \sim$ Bernoulli(1/2), and $N = \inf\{n : X_{n+1} = 1\}$. By definition, $\mathbb{E}[\sum_{i=1}^N X_i] = 0$. However, $\mathbb{E}[X_1] = 1/2$, and $\mathbb{E}[N + 1] = 1(1/2) + 2(1/2)^2 + 3(1/2)^3 + \ldots = \sum_{k=1}^{\infty} k2^{-k} > 1$, so $\mathbb{E}[X_1]\mathbb{E}[N] \neq 0$!

One example when stopping times play a special role is in Wald's identities.

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Theorem 12.2 (Wald's identity)**.** *Let $X_1, X_2, \dots$ be i.i.d. random variables and N be a random time. If $X_i \geq 0$ and N is a stopping time, then*

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \mathbb{E}[X_1] \cdot \mathbb{E}[N].$$

Proof. Note that we can write

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \mathbb{E}\left[\sum_{i=1}^{\infty} X_i \mathbb{1}_{\{N \geq i\}}\right] = \sum_{i=1}^{\infty} \mathbb{E}\left[X_i \mathbb{1}_{\{N \geq i\}}\right].$$

For the final equality, we used the assumption that $X_i \geq 0$ to interchange the order of integration by Fubini-Tonelli's theorem. Continuing, by using the assumption that N is a stopping time (so that $\mathbb{1}_{\{N \geq i\}} = (\mathbb{1}_{\{N \leq i-1\}})^c$ is $\sigma(X_1, \dots, X_{i-1})$-measurable), and since X_i is independent of $\sigma(X_1, \dots, X_{i-1})$ and $\mathbb{E}[X_i] = \mathbb{E}[X_1]$,

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \sum_{i=1}^{\infty} \mathbb{E}[X_i]\mathbb{P}(N \geq i) = \mathbb{E}[X_1]\mathbb{E}[N].$$

□

Remark 12.3.
- Observe that the "tail formula" for the expectation of a $\mathbb{N}_+$-valued random variable X:
$$\sum_{i=1}^{\infty} \mathbb{P}(X \geq i) = \mathbb{E}(X)$$
is just an instance of the integration by parts formula (Lecture 6).
- Note that nonnegativity of X_i was only used to apply Fubini's theorem, and actually the condition that the stochastic process and and the stopping time function need to satisfy is
$$\sum_{i=1}^{\infty} \mathbb{E}\left[|X_i| \mathbb{1}_{\{N \geq i\}}\right] < \infty$$
(again, Lecture 6). This is important for the next example.

Example 12.4 (**Gambler's ruin**)**.** *Suppose that we bet a dollar on the outcome of a fair coin flip and win one dollar if it comes up heads (with probability $1/2$), and lose the dollar if it comes up tails. The game stops when we lose all our money, or when we win big (say, M dollars). What is the probability of winning?*

Let $X_1, X_2, \dots$ be i.i.d. Rademacher random variables such that $X_1 = \pm 1$ with probability $1/2$ modelling the outcome of each coin toss. Let $S_n = \sum_{i=1}^{n} X_i$ denote the winnings after n coin tosses, and consider the stopping time

$$T = \inf\{n : S_n = M \text{ or } S_n = -m\}.$$

Observe that $\mathbb{E}[T] < (M+m)2^{M+m} < \infty$. This loose upper bound can be argued by saying that T is dominated by a stopping time for when we first observe $M+m$ consecutive wins. However, an exact formula can actually be derived for $\mathbb{E}[T]$ in this problem.

Note that we can apply Wald's identity. Indeed,

$$\sum_{i=1}^{\infty} \mathbb{E}\left[|X_i| \mathbb{1}_{\{T \geq i\}}\right] = \sum_{i=1}^{\infty} \mathbb{E}\left[\mathbb{1}_{\{T \geq i\}}\right] = \sum_{i=1}^{\infty} \mathbb{P}(T \geq i) = \mathbb{E}T < \infty$$

which is enough by Remark 12.3.

By Wald's identity,

$$\mathbb{E}[S_T] = \mathbb{E}[T] \cdot \mathbb{E}[X_1] = 0$$

since clearly $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[T] < \infty$ as checked above.

Atthe same time, we can compute this expectation by definition:

$$\mathbb{E}[S_T] = M\mathbb{P}(\text{win}) - m\mathbb{P}(\text{lose}) = 0.$$

Since $\mathbb{P}(\text{lose}) = 1 - \mathbb{P}(\text{win})$, this implies that $M\mathbb{P}(\text{win}) - m + m\mathbb{P}(\text{win}) = 0$, so that by rearranging,

$$\mathbb{P}(\text{win}) = \frac{m}{M+m}.$$

Think through why the answer agrees with the intuitively expected one. What does it mean $m \ll M$, what happens if you are allowed to lose "infinite" amount of money and keep playing and what makes the probabilities to win and to lose equal?

Optional stopping theorem

Key question: is $\mathbb{E}[X_\tau] = \mathbb{E}[X_0]$ for a martingale?

It is true for the martingale $S_n = \sum_{i=1}^n X_i$ when X_i are i.i.d. and mean zero, and $\mathbb{E}[T] < \infty$ (Wald's identity). We can immediately see that some assumptions on T are needed:

Example 12.5. Consider $S_0 = 0$, $S_n = X_1 + \ldots + X_n$ where X_i are i.i.d. Rademacher random variables (i.e., $X_i = \pm 1$ with equal probability), and the stopping time

$$\tau = \inf\{n \geq 1 : S_n = 1\}.$$

Then we have the following:

- $\tau < \infty$ a.s.: this follows from a combinatorial fact: for all $n \geq 0$,
$$\mathbb{P}(\tau = 2n+1) = \frac{1}{2^{n+1}} C_n,$$
where $C_n = \frac{1}{n+1}\binom{2n}{n}$ is the nth Catalan number. Thus, $\mathbb{P}(\tau < \infty) = 1$.
- $\mathbb{E}[S_0] = 0$, but $\mathbb{E}[S_\tau] = 1$!

Question: what is $\mathbb{E}[T] =$? Note that the reasoning from Gambler's ruin does not apply as the hitting set $\{1\}$ does not have both lower and upper bound (so, no finite sequence of heads guarantees hitting it).

Definition 12.6 (Stopped process)**.** Given a stochastic process $(X_n)_{n\geq 0}$ and a stopping time τ, we define the **stopped process** $X^\tau := (X_{\tau\wedge n})_{n\geq 0}$. (Here $\tau \wedge n = \min\{\tau, n\}$.)

Theorem 12.7. *If $(X_n)_{n\geq 0}$ is a martingale/submartingale/supermartingale and τ is a stopping time, then $X^\tau = (X_{\tau\wedge n})_{n\geq 0}$ is a martingale/submartingale/supermartingale respectively.*

Proof. Suppose that $(X_n)_{n\geq 0}$ is a martingale. Let $A_n := \mathbb{1}_{\{\tau\geq n\}}$. The key observation is that A_n is $\mathcal{F}_{n-1}$-measurable (and therefore defines a predictable process), bounded and nonnegative. Therefore, by Lemma 11.5,

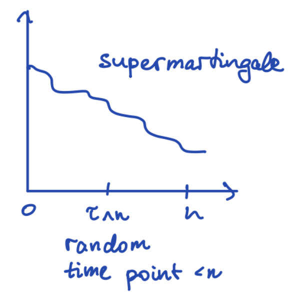
$$(A \bullet X)_n = \sum_{k=1}^n A_k(X_k - X_{k-1}) = \sum_{k=1}^n \mathbb{1}_{\{\tau\geq k\}}(X_k - X_{k-1}) = \sum_{k=1}^{\tau\wedge n}(X_k - X_{k-1}) = X_{\tau\wedge n} - X_0$$

is a martingale. The proof for when X_n is a submartingale/supermartingale is similar (and it is also easy to check directly). ☐

Corollary 12.8 (Doob's inequality)**.** *If $(X_n)_{n\geq 0}$ is a supermartingale, then*

$$\mathbb{E}X_0 \geq \mathbb{E}X_{\tau\wedge n} \geq \mathbb{E}X_n.$$

Note that if X_n is a martingale then $\mathbb{E}X_n = \mathbb{E}X_0$ so we have the above holds with the equalities.



Proof. Since $X_{\tau\wedge n}$ is a supermartingale by Theorem 12.7, we have $\mathbb{E}[X_{\tau\wedge n}] \leq \mathbb{E}[X_{\tau\wedge 0}] = \mathbb{E}[X_0]$. Next, for $k \leq n$, since $\mathbb{1}_{\{\tau=k\}}$ is $\mathcal{F}_k$-measurable, we have the inequality

$$\mathbb{E}[X_k \mathbb{1}_{\{\tau=k\}}] \geq \mathbb{E}[\mathbb{E}[X_n \mid \mathcal{F}_k]\mathbb{1}_{\{\tau=k\}}] = \mathbb{E}[X_n \mathbb{1}_{\{\tau=k\}}].$$

So,

$$\begin{aligned}\mathbb{E}[X_{\tau\wedge n}] &= \sum_{k=0}^{n} \mathbb{E}[X_k \mathbb{1}_{\{\tau=k\}}] + \mathbb{E}[X_n \mathbb{1}_{\{\tau>n\}}] \\ &\geq \sum_{k=0}^{n} \mathbb{E}[X_n \mathbb{1}_{\{\tau=k\}}] + \mathbb{E}[X_n \mathbb{1}_{\{\tau>n\}}] = \mathbb{E}[X_n].\end{aligned}$$
☐

Based on what we just proved, the following might make sense for a martingale X_n: $\mathbb{E}[X_{\tau\wedge n}] = \mathbb{E}[X_{0\wedge n}] = \mathbb{E}[X_0]$. If $\tau < \infty$ a.s., then we can send $n \to \infty$ to obtain $X_{\tau\wedge n} \to X_\tau$. *However, it is not always true!* Recall Example 12.5. Thus, we need to impose some sort of integrability conditions.

Theorem 12.9 (Doob's optional stopping theorem)**.** *Let $(X_n)_{n\geq 0}$ be a martingale, and suppose that τ is a stopping time such that $\tau < \infty$ a.s., and that one of the following holds:*

(a) *(τ is bounded). There exists a constant $K < \infty$ such that $\tau \leq K$ a.s.*
(b) *($X_{\tau\wedge n}$ is bounded). There exists a constant $K < \infty$ such that $|X_{\tau\wedge n}| \leq K$ a.s.*
(c) *(τ has finite expectation and $X_{\tau\wedge n}$ has bounded increments). $\mathbb{E}[\tau] < \infty$, and there exists a constant $K < \infty$ such that $|X_{\tau\wedge(n+1)} - X_{\tau\wedge n}| \leq K$ for all n a.s.*

Then $\mathbb{E}[X_\tau] = \mathbb{E}[X_0]$.

Proof. (a) Since $\tau \leq K$ a.s., $\mathbb{E}[X_\tau] = \mathbb{E}[X_{\tau\wedge K}] = \mathbb{E}[X_0]$, since $X_{\tau\wedge n}$ is a martingale by Theorem 12.7.

For the other parts, we aim to apply the DCT. That is, since $X_{\tau\wedge n} \to X_\tau$ as $n \to \infty$, if we can show that $|X_{\tau\wedge n}|$ is dominated by an integrable function, we may take the limit as $n \to \infty$ in the identity $\mathbb{E}[X_{\tau\wedge n}] = \mathbb{E}[X_0]$ and use the DCT to deduce that $\mathbb{E}[X_\tau] = \mathbb{E}[X_0]$.

(b) We can simply take the constant K to be the integrable dominating function.

(c) Note that we can write $X_{\tau\wedge n} = X_0 + \sum_{i=1}^{\tau\wedge n}(X_{\tau\wedge i} - X_{\tau\wedge(i-1)})$. Therefore, since the increments are bounded by K,

$$|X_{\tau\wedge n}| \le |X_0| + \sum_{i=1}^{\tau\wedge n} |X_{\tau\wedge i} - X_{\tau\wedge(i-1)}| \le |X_0| + \tau K.$$

Taking expectations, we have $\mathbb{E}|X_{\tau\wedge n}| \le \mathbb{E}|X_0| + K\,\mathbb{E}[\tau] < \infty$ by assumption. □

Example 12.10 (**ABRACADABRA**). *Suppose that a monkey types a letter from the twenty six characters of the English alphabet uniformly at random in each second. How long will it take the monkey to type the string ABRACADABRA? More precisely, consider the stopping time*

$$T := \inf\{t \ge 0 : \textit{ABRACADABRA typed in } t \textit{ seconds}\}.$$

Then what is $\mathbb{E}[T]$*?*

Firstly, we can show that $\mathbb{E}[T] < \infty$ by providing a loose upper bound via the following argument. Note that in any block of 11 characters (i.e., the length of the desired string), the probability of success is $1/26^{11}$. Therefore, if we consider chopping up time into consecutive blocks of size 11, T is dominated by the first time in which ABRACADBRA appears in any of these blocks of 11 characters, so $\mathbb{E}[T] \le 11\cdot 26^{11} < \infty$ (since the expected number of blocks is given by the expectation of a Geometric($1/26^{11}$) random variable, which is 26^{11}). Is this tight? Not really (e.g., the phrase can appear in between two of these consecutive blocks). In fact, we will be able to compute $\mathbb{E}[T]$ *exactly* by constructing a relevant martingale!

Consider the following fair game. At each time point t, a new gambler arrives with one dollar and makes bets on the character that the monkey will type in time t and afterwards according to the following strategy:

- They bet all their money on "A": if they lose, they leave; otherwise, they win \$26 and continue to gamble. (This amount makes the game fair, since the expected winnings is zero.)
- They bet all their money on "B": if they lose, they leave; otherwise, they win \$$26^2$ and continue to gamble.
- Repeat until "ABRACADABRA" appears, after which the gambler, if they have not already left, happily collects their final earnings (of \$$26^{11}$) and leaves.

Let X_n denote the increase in wealth of the casino at time n from the inflow of gamblers. That is,

$$X_n = n - (\text{money currently paid out to gamblers still in the game}).$$

Let us check that X_n is a martingale! Indeed,

$$\begin{aligned}\mathbb{E}[X_n|\mathcal{F}_{n-1}] &= \mathbb{E}[(n-1) + 1 - (\text{money that gamblers hold at time } n)|\mathcal{F}_{n-1}]\\ &= \mathbb{E}[X_{n-1} + 1 + (\text{money gamblers hold at } n-1) - (\text{money gamblers hold at } n)|\mathcal{F}_{n-1}]\\ &= X_{n-1} + 1 - \mathbb{E}[(\text{player gains at round } n) + 1|\mathcal{F}_{n-1}] = X_{n-1}.\end{aligned}$$

Here, $+1$ in the last line is a dollar contributed by a new player at time n; expected gains of all players are 0 since they are 0 for every player:

$$\mathbb{E}(\text{gain}) = 1/26 * (26(\text{bet}) - (\text{bet})) - 25/26 * (\text{bet}) = 0.$$

Note that $\mathbb{E}[T] < \infty$ by our initial estimate, and that the increments of $X_{T\wedge n}$ are bounded by at most 26^{11} (i.e., the biggest payout to a lucky gambler who witnesses the realization of ABRACADABRA). Thus, by the optional stopping theorem (Theorem 12.9), $\mathbb{E}[X_T] = \mathbb{E}[X_0] = 0$. Since

$$\mathbb{E}[X_T] = \mathbb{E}[T] - 26^{11} - 26^4 - 26 = 0,$$

(the amounts 26^{11}, 26^4, and 26 correspond to the gamblers who are collecting their earnings for seeing "ABRACADABRA", "ABRA", and "A" respectively), we conclude that

$$\mathbb{E}[T] = 26^{11} + 26^4 + 26 \approx 3.67 \times 10^{15}.$$

Optimal stopping of general processes

Question: how can martingale-based techniques help us to stop a stochastic process optimally? (Supermartingale covers.)

Consider the following setup for the so-called finite horizon stopping problem:

- We are given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a filtration $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \ldots \subseteq \mathcal{F}_N$ over a finite horizon.
- Let $X_1, \ldots, X_N$ be integrable random variables, *not necessarily adapted to the filtration.* (Interpretation: X_n is the reward at time n, and $\mathcal{F}_n$ is the information available at time n; reward is not fully known when you can take it.)
- Can we find a stopping time that maximizes the expected reward $\mathbb{E}[X_\tau]$?

Snell envelope technique: consider adapting the process to the filtration, covering it with a supermartingale, and stopping as soon as the adapted process dominates the cover. Specifically, consider the following construction:

(1) $Y_n = \mathbb{E}[X_n \mid \mathcal{F}_n]$.
(2) $V_n = \max\{Y_n, \mathbb{E}[V_{n+1} \mid \mathcal{F}_n]\}$, defined by backward induction from $V_N = Y_N$. ($V_n =$ Snell envelope.)
(3) $\tau = \min\{n : Y_n = V_n\}$.

Remark 12.11. It can be directly checked from the definition that V_n is a supermartingale that upper bounds Y_n. It can also be shown that V_n is the "smallest" supermartingale that dominates Y_n, which explains why it is called an "envelope" (see [Chatterjee, Exercise 9.8.2]).

Theorem 12.12. *Consider the Snell envelope V_n and stopping time τ defined above. If T is any stopping time adapted to $\mathcal{F}_n$, then*

$$\mathbb{E}[X_T] \leq \mathbb{E}[X_\tau] = \mathbb{E}[V_1].$$

Before proving this theorem, we state a proposition that generalizes the result that a stopped martingale is a martingale (Theorem 12.7). For a proof of this result, see [Chatterjee, Proposition 9.7.2].

Proposition 12.13. *Let $(X_n)_{n\geq 0}$ be a sequence of integrable random variables adapted to a filtration $(\mathcal{F}_n)_{n\geq 0}$, and T be a stopping time for this filtration. If $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ a.s. on the event $\{T > n\}$, then $X_{T\wedge n}$ is a martingale.*

We will now prove Theorem 12.12 on the optimal stopping time.

Proof of Theorem 12.12. First, note that for any stopping time T, $\mathbb{E}[X_T] = \mathbb{E}[Y_T]$. This follows from the equality

$$\mathbb{E}[Y_T] = \sum_{n=1}^{N} \mathbb{E}[Y_n \mathbb{1}_{\{T=n\}}] = \sum_{n=1}^{N} \mathbb{E}[X_n \mathbb{1}_{\{T=n\}}] = \mathbb{E}[X_T],$$

where the second equality uses the fact that $Y_n = \mathbb{E}[X_n \mid \mathcal{F}_n]$ and $\{T = n\} \in \mathcal{F}_n$.

Note that on the set $\{\tau > n\}$, $V_n = \mathbb{E}[V_{n+1} \mid \mathcal{F}_n]$ a.s. Therefore, by Proposition 12.13, $(V_{\tau\wedge n})_{1\le n\le N}$ is a martingale adapted to $(\mathcal{F}_n)_{1\le n\le N}$. Moreover, $Y_\tau = V_\tau$ by the construction of τ. Hence, using the fact that $1 \wedge \tau = 1$ and $N \wedge \tau = \tau$, we have

$$\mathbb{E}[V_1] = \mathbb{E}[V_{1\wedge\tau}] = \mathbb{E}[V_{N\wedge\tau}] = \mathbb{E}[V_\tau] = \mathbb{E}[Y_\tau] = \mathbb{E}[X_\tau].$$

At the same time, since V_n is a supermartingale that upper bounds Y_n, and T is a bounded stopping time (any stopping time is bounded as we do finitely many steps until N) we have by the Optional Stopping Theorem that

$$\mathbb{E}[V_1] \ge \mathbb{E}[V_T] \ge \mathbb{E}[Y_T] = \mathbb{E}[X_T]$$

Combining the two preceding displayed equations completes the proof. □

Princess/secretary problem

Suppose that we have a pool of N candidates with ranks $(1,\dots,N)$ from most preferred to least preferred. They are interviewed sequentially in a random order. Each candidate is either chosen and the process stops, or they are dismissed (and dismissed candidates cannot be recalled). At time $1 \le n \le N$, only the relative ranks of the first n candidates $(1,\dots,n)$ are observed. ***Question: when is the "best" time to stop?*** *More precisely, suppose that the "best" time maximizes the probability that the chosen candidate is the most preferred (there are variants of this problem for different notions of "best").*

Let us reformulate this problem to apply the Snell envelope technique. Let $X_n = \mathbb{1}_{\{r_n=1\}}$, where r_n denotes the absolute rank of candidate n. Let $\mathcal{F}_n$ denote the relative order of the candidates revealed by time n. What is $\max_T \mathbb{E}[\mathbb{1}_{\{r_T=1\}}]$ over stopping times T?

We know from the general theory above that for these rewards X_n and filtration $\mathcal{F}_n$, if we define $Y_n = \mathbb{E}[X_n \mid \mathcal{F}_n]$, $V_n = \max\{Y_n, \mathbb{E}[V_{n+1} \mid \mathcal{F}_n]\}$ for $n = N-1, N_2, \dots$ with $V_N = Y_N$, and $\tau = \min\{n : Y_n = V_n\}$, then $\mathbb{E}[X_\tau] = \mathbb{E}[V_1]$ maximizes the expectation for any other stopping time T.

Goals:

(a) Define τ explicitly for this problem.
(b) Estimate τ and the expected reward $\mathbb{E}[X_\tau]$ (probability of successful choice of top candidate).

First, let's compute $Y_n = \mathbb{P}(r_n = 1 \mid \text{relative order observed by time } n)$. If the relative order of the nth candidate is not the top at time n, then $Y_n = 0$. Otherwise,

$$\mathbb{P}(r_n = 1 \mid \text{observed order with } n \text{ top at time } n)$$

$$= \frac{\mathbb{P}(\text{observed order with } n \text{ top at time } n \mid r_n = 1)\cdot\mathbb{P}(r_n = 1)}{\mathbb{P}(\text{observed order with } n \text{ top at time } n)} = \frac{\frac{1}{(n-1)!}\cdot\frac{1}{N}}{\frac{1}{n!}} = \frac{n}{N}.$$

So,

$$Y_n = \frac{n}{N} \cdot \mathbb{1}_{\{n\text{th observed as relatively best}\}}.$$

Note that Y_n only takes two values 0 and n/N. Since there are $n!$ relative orders of the first n candidates, and $(n-1)!$ relative orders of the other $n-1$ candidates if we fix the nth candidate at the top of the relative order at time n, we have

$$\begin{aligned}
&\mathbb{P}(Y_n = n/N \mid \text{relative order of first } n-1 \text{ candidates}) \\
&\quad= \mathbb{P}(n\text{th observed as relatively best} \mid \text{relative order of first } n-1 \text{ candidates}) \\
&\quad= \frac{\mathbb{P}(\text{relative order of first } n \text{ candidates})}{\mathbb{P}(\text{relative order of first } n-1 \text{ candidates})} = \frac{\frac{1}{n!}}{\frac{1}{(n-1)!}} = \frac{1}{n}.
\end{aligned}$$

Thus, $\mathbb{P}(Y_n = n/N) = \mathbb{P}(Y_n = n/N \mid \mathcal{F}_{n-1}) = \frac{1}{n}$, which shows that Y_n is independent of $\mathcal{F}_{n-1}$ (i.e., the relative order of the first $n-1$ candidates is independent from the rank of the next candidate).

Next, let's discuss $V_n = \max\{Y_n, \mathbb{E}[V_{n+1} \mid \mathcal{F}_n]\}$, defined backwards with $V_N = Y_N$. We can show that $\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] = v_n$ for some constant (non-random) quantity v_n for all $1 \leq n \leq N-1$ by backwards induction. Indeed, since $V_N := Y_N$ and Y_N is independent of $\mathcal{F}_{N-1}$ from the previous calculations, we have

$$\mathbb{E}[V_N \mid \mathcal{F}_{N-1}] = \mathbb{E}[Y_N \mid \mathcal{F}_{N-1}] = \mathbb{E}[Y_N] = \frac{1}{N} =: v_N.$$

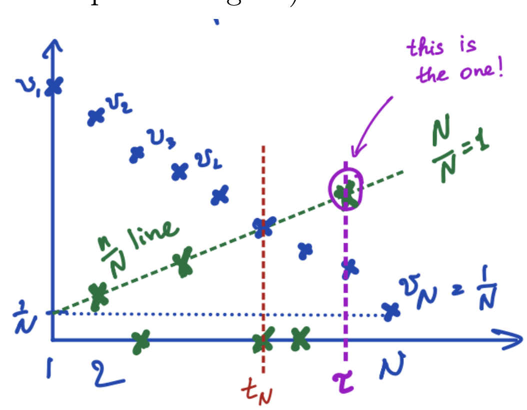
Thus, the claim holds for $n = N-1$. If the claim holds for some n, then

$$\begin{aligned}
\mathbb{E}[V_n \mid \mathcal{F}_{n-1}] &= \mathbb{E}[\max\{Y_n, \mathbb{E}[V_{n+1} \mid \mathcal{F}_n]\} \mid \mathcal{F}_{n-1}] \\
&= \mathbb{E}[\max\{Y_n, v_n\} \mid \mathcal{F}_{n-1}] = \mathbb{E}[\max\{Y_n, v_n\}] =: v_{n-1},
\end{aligned}$$

where we used the fact that Y_n is independent of $\mathcal{F}_{n-1}$ and v_n is non-random by the induction hypothesis. Thus, the claim holds for all n. From this proof, we have seen that $v_N = 1/N$, and since $v_{n-1} = \mathbb{E}[\max\{Y_n, v_n\}]$, we have

$$v_1 \geq v_2 \geq \ldots \geq v_N = \frac{1}{N}.$$

(That is, the Snell envelope is a supermartingale!)



Finally, we can now discuss $\tau = \min\{n : Y_n = V_n\}$. We have shown that the values of Y_n are random but jump between 0 and n/N, and v_n are decreasing constants from some value to $1/N$. Since $V_n = \max\{Y_n, v_n\}$, from our calculations above, τ stops the process when the values of Y_n first dominate v_n. Note that this must occur at a time $n \geq t_N$, where

$$t_N := \min\{n \leq N-1 : n/N \geq v_n\},$$

since for any earlier time $n < t_N$, $Y_n \leq n/N < v_n$, and therefore $v_{n-1} = \mathbb{E}[\max\{Y_n, v_n\}] = v_n$. Hence, by Theorem 12.12, we can explicitly write the optimal stopping time as

$$\begin{aligned}\tau &= \min\{n \geq t_N : Y_n = n/N\} \\ &= \min\{n \geq t_N : n\text{th candidate observed as relatively best}\},\end{aligned}$$

such that $\max_T \mathbb{E}[X_T] = \mathbb{E}[X_\tau] = \mathbb{E}[V_1] = \mathbb{E}[V_{t_N-1}]$.

It remains to estimate when v_n first passes the n/N threshold. By definition and our computations above, we have $v_N = 1/N$, and for $n \geq t_N$,

$$v_{n-1} = \frac{n}{N}\mathbb{P}(Y_n = n/N) + v_n \mathbb{P}(Y_n = 0) = \frac{1}{N} + \left(1 - \frac{1}{n}\right) v_n.$$

Hence, by backward induction, it can be shown that for $t_N - 1 \leq n \leq N-1$,

$$v_n = \frac{1}{N} + \frac{n}{N}\sum_{k=n+1}^{N-1} \frac{1}{k}.$$

Thus, t_N is the unique integer $n \in \{1, 2, \ldots, N-1\}$ such that

$$\frac{1}{N} + \frac{n}{N}\sum_{k=n}^{N-1}\frac{1}{k} > \frac{n}{N} \geq \frac{1}{N} + \frac{n}{N}\sum_{k=n+1}^{N-1}\frac{1}{k}.$$

For our final goal, it remains to estimate this quantity. Say $t_N = n \sim \alpha N$ for some parameter factor α (let's find it!). Then the equation that must jold is

$$\begin{aligned}\alpha \approx \frac{1}{N} + \frac{n}{N}\sum_{k=n+1}^{N-1}\frac{1}{k} &= 0 + \alpha \sum_{k=\lfloor \alpha N \rfloor}^{N} \frac{1}{k} \\ &\approx \alpha \int_{\alpha N}^{N} \frac{1}{x}\,\mathrm{d}x = \alpha(\log N - \log(\alpha N)).\end{aligned}$$

This implies

$$1 \approx \log N - \log(\alpha N) = \log\left(\frac{N}{\alpha N}\right) = \log(\alpha^{-1}), \quad \text{and so} \quad 1/\alpha \approx e.$$

To summarize, this means that the optimal strategy is to wait and observe the first N/e candidates (out of the total N candidates; or, a $1/e \approx 36.79\%$ proportion) without making a decision. After that, make an offer to the first candidate who is better than all the candidates that have been previously observed. This strategy maximizes the expected reward (being the probability that the selected candidate is the top ranked) over all stopping times, which is equal to $\mathbb{E}[X_\tau] = \mathbb{E}[V_1] = \mathbb{E}[V_{t_N-1}] \sim 1/e$.

Finally, note that compared to the naive strategy of selecting the first or the last candidate, $1/e \gg 1/N$.